

Generation Interconnection System Impact Study Report

For Queue Project Q14-0348

Mount Laurel 138 kV

September 27, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0348. The Interconnection Customer proposes a 105.5 MW natural gas combined cycle facility interconnecting at the Mount Laurel 138 kV substation in the New Jersey area. The request was received on June 27, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 212 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$372,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WALSINGHAM / CORINTH 138.0 kV
Contingency: LOLO SUBSTATION / CORINTH 138.0 kV Line
Loading: 109.5% (Rating: 145 MVA)
DFAX: 8.3%
Pre-project loading: 103.4%
2. Facility: CORINTH / PHILO SUBSTATION 138.0 kV
Contingency: OAKLAND X / MOUNT LAUREL 138.0 kV
Loading: 103.8% (Rating: 386 MVA)
DFAX: 24.1%
Pre-project loading: 97.2%
3. Facility: MOUNT LAUREL / PHILO 138.0 kV
Contingency: OAKLAND X / MOUNT LAUREL 138.0 kV L
Loading: 102.5% (Rating: 421 MVA)
DFAX: 24.1%
Pre-project loading: 96.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WALSINGHAM / CORINTH 138.0 kV	Reconductor line	\$1,584,000	8% (\$131,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CORINTH / PHILO SUBSTATION 138.0 kV	Reconductor line	\$278,000	9% (\$26,000)
MOUNT LAUREL / PHILO 138.0 kV	Reconductor line	\$2,332,000	9% (\$215,000)
	Total allocated to project		\$372,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.